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The Labor Market Effects of Immigration Enforcement

Chloe N. East, Annie L. Hines, Philip Luck, Hani Mansour, Andrea Velásquez

Online Appendix

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I. Data Description

A. CZ-Year Control Variables

A.1 Bartik-Style Measures of Local Labor Demand

The Bartik measures predict labor demand for each group by taking advantage of the fact that different demographic groups work in different industries, and that this varies across local areas. This is done with a shift-share approach as follows:

$$Bartik_{jt} = \sum_{i=1}^{20} \frac{emp_{ij2005}}{emp_{j2005}} * emp_{it} * \frac{1}{100,000} \quad (1)$$

We use the 2005 ACS to create the "share" measures ($\frac{emp_{ij2005}}{emp_{j2005}}$); this is the share of group-specific CZ j employment in 2005 that is in industry i. The reason we use the 2005 ACS instead of, say the 2000 Census, is that the geographic codes are different in the Census, so this allows us to create variables that match the geography in our main analysis. We use 20 large industry groups for these calculations. We calculate the national shift measures (emp_{it}) – total number of workers in industry i and year t – with the 2005-2014 ACS. We divide by 100,000 to make the presentation of the results easier.

A.2 Housing Price and Start Information

The housing price information used in the trend control in Tables (7) and (8) comes from the Federal Housing Finance Agency and is available at the county by year level. Similarly, the housing permit ("start") information is available from the U.S. Census Bureau at the county by year level. We aggregate both housing data sets up to the CZ level using the weighting process as described in the main text for the SC variable.

A.3 287(g) Agreements

We also include controls for the presence of local-level 287(g) agreements in Tables (7) and (8). 287(g) agreements were similar to SC, but 287(g)s were optional agreements law en-

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forcement agencies could choose to enter into with the federal government. There were three types of 287(g) agreements, and we control for the types separately. The "Task Force" model permitted trained law enforcement officials to screen individuals regarding their immigration status during policing operations, and arrest individuals due to suspected immigration violations. The "Jail" model allowed screening of immigration status for individuals upon being booked in state prisons or local jails and was more like SC. A third "Hybrid" model includes both the Task Force and Jail models.[1] In 2012, due to criticism about the efficacy of the program, including issues of racial profiling, Task Force agreements were no longer renewed. Start and end dates for all 287(g) agreements came from reports published by ICE, the Department of Homeland Security, the Migration Policy Institute, as well as Kostandini et al. (2013), and various news articles.

A.4 E-Verify

We control for the presence of state-level E-Verify policies in Tables (7) and (8). E-Verify is an employment-based immigration enforcement program which required some employers to verify eligibility for work in the U.S. of job applicants through a federal database. Some states only required government employers and government-contracted employers to do this, whereas other states required all private employers over a given size to do so. We control for the presence of each type of E-Verify program (public only vs. all) separately. Since many CZs span multiple states, we take the population-weighted average value of the E-Verify variables across states within a CZ-year.

B. TRAC Data Description

Data on deportations under SC comes from the Transactional Records Access Clearinghouse at Syracuse University. TRAC obtained these data from ICE through a series of Freedom of Information Act requests. The data contain individual-level records of each deportation

[1] Background information on 287(g)s is obtained from Capps et al. (2011).

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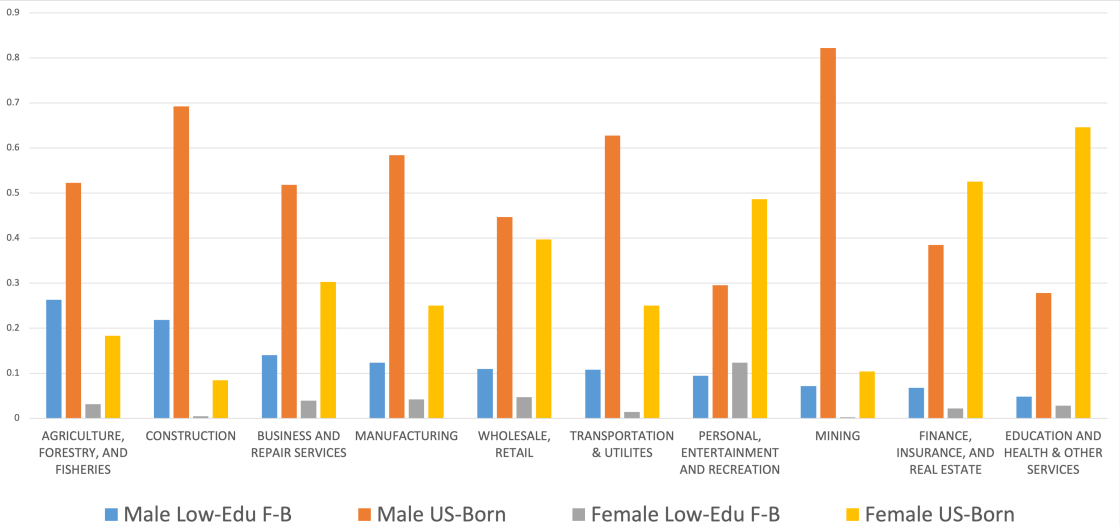
under SC, beginning in November 2008 and continuing through the end of SC in 2014.[2] The county given in this file is the county of apprehension, the date is the date of removal. Because deportations do not happen immediately upon apprehension, there is a lag between the initial apprehension and the date recorded in our data. For each individual, we have information on the deportation proceedings as well as various demographics, including age, gender, and country of citizenship. The data also contain information on the criminal background of the deportee, including their most serious criminal conviction (MSCC).

TRAC provides a similar file of records for ICE detainers, which we use to examine the effects of SC on detention intensity. However, it is important to note that we cannot separately identify which detentions were done under SC.

[2]The data also contain information about deportations under PEP, which replaced SC in 2014, as well as under the restoration of SC after January 2017, but we focus our attention on the first period SC was in place: 2008-2014.

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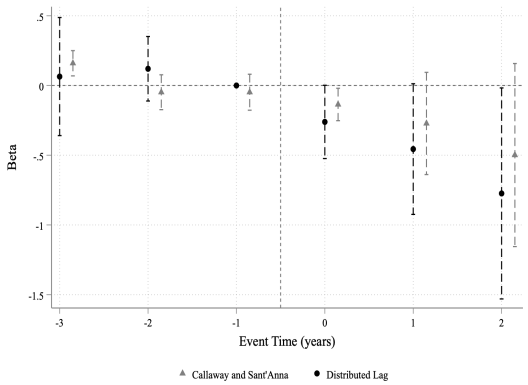
Figure A1: Share of Sector Workers in 2005 that are Low-Educated Foreign-Born and U.S.-Born, Split by Sex



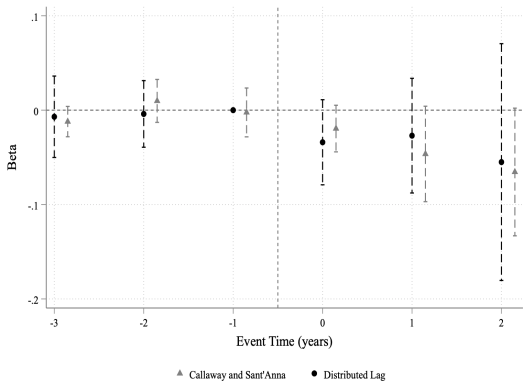
Notes: Data are drawn from the 2005 American Community Survey. The sample includes all working-age individuals who report an industry of current or recent employment. The bars show the share of total sector employment made up by each demographic group: male low-educated foreign-born, male U.S.-born, female low-educated foreign-born, and female U.S.-born, respectively. The results are weighted using the ACS-provided person weights.

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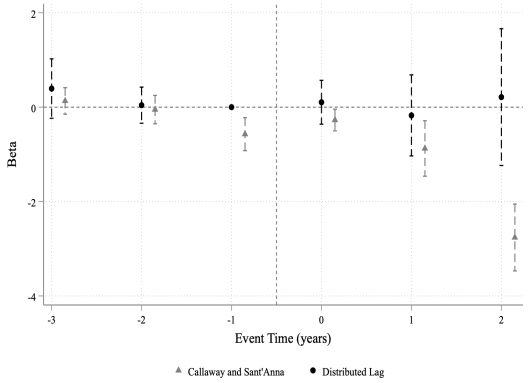
Figure A2: Effects of SC on the Employment Share and Wages of Low-Educated Foreign-Born and U.S.-Born, using Callaway and Sant'Anna (2020)



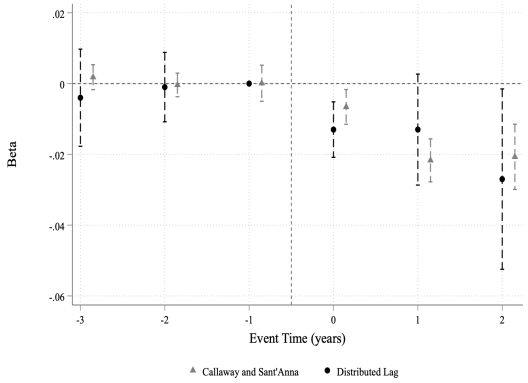
Panel A: Low-Educated Foreign-Born Employment Share



Panel B: Low-Educated Foreign-Born Wages



Panel C: U.S.-Born Employment Share

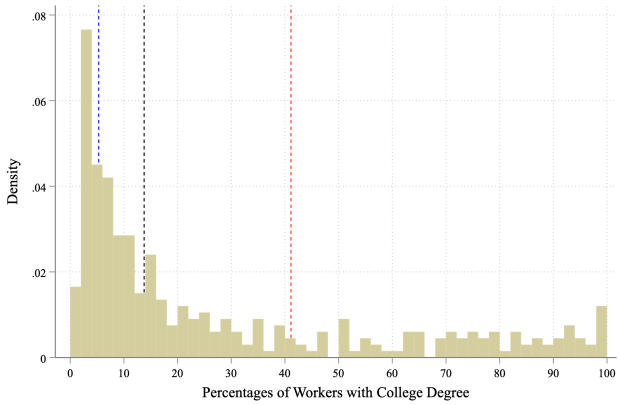


Panel D: U.S.-Born Wages

Notes: Data are drawn from the 2005-2014 American Community Survey. The samples are based on working-age (20-64) low-educated (high school or less) foreign-born individuals and U.S.-born individuals. In Panels A and C, the employment share is measured as total employment in the demographic group by CZ and year, divided by total working-age CZ Population in 2005, and then multiplied by 100. In Panels B and D, hourly wages are calculated by dividing annual earnings by annual hours worked among full-time workers. Distributed lag model γ_k coefficients are estimated using equation (2) and then transformed into total treatment effects ($\beta_k s$), which are plotted in the solid black dots. The 95% confidence intervals of the $\beta_k s$ are shown in the black dashed lines. The distributed lag models include CZ fixed effects, year fixed effects, and CZ-specific linear trends. The Callaway and Sant'Anna estimates are based on de-trended data, and are plotted with gray triangles. The corresponding 95% confidence intervals are shown by the gray dashed lines. All results are weighted using the CZ population in 2000 and all standard errors are clustered by CZ.

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Figure A3: Distribution of Workers with a College Degree or More Across Occupations



Notes: The figure plots the density of our occupational skill measure–the fraction of workers in each occupation with at least a college degree. This is calculated using the 2005 American Community Survey. The black bar indicates the occupation with the median skill (13.78 percent of workers in this median skill have a college degree), and the blue and red bars depict the 25th and 75th percentile skill occupations, respectively (5.30 and 41.13 percent of workers in these skills have a college degree, respectively).

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Table A1

Summary Statistics

Employment Share of Low-Educated Foreign-Born	
Low-Educated Foreign-Born (LEFB)	6.75
LEFB Female Individuals	2.55
LEFB Male Individuals	4.21
Employment Share of U.S.-Born	
All	58.61
Female Individuals	28.39
Male Individuals	30.22
Hourly Wages of Low-Educated Foreign-Born	
Low-Educated Foreign-Born (LEFB)	15.60
LEFB Female Individuals	13.60
LEFB Male Individuals	16.79
Hourly Wages of U.S.-Born	
All	24.39
Female Individuals	21.51
Male Individuals	26.91

Notes: Data are drawn from the 2005-2014 American Community Survey. The low-educated foreign-born sample includes working-age (20-64) individuals with a high school degree or less. The U.S.-born sample includes working-age (20-64) individuals. The employment share is measured as total employment in the demographic group by CZ and year, divided by total working-age CZ Population in 2005, and then multiplied by 100. Hourly wages are calculated by dividing annual earnings by annual hours worked among full-time workers. See section 3 for more details. Summary statistics are weighted by the CZ population in 2000.

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Table A2
Distributed Lag Model Coefficients

	Dep. Var: Total Group Emp / Total CZ Pop in 2005 * 100		Dep. Var: Log Hourly Wages	
	Low-Edu Foreign-Born	US-Born	Low-Edu Foreign-Born	US-Born
-3	0.064	0.394	-0.007	-0.004
	(0.216)	(0.321)	(0.022)	(0.007)
-2	0.120	0.042	-0.004	-0.001
	(0.118)	(0.195)	(0.018)	(0.005)
-1	omitted	omitted	omitted	omitted
0	-0.261	0.104	-0.034	-0.013
	(0.134)	(0.237)	(0.023)	(0.004)
1	-0.456	-0.175	-0.027	-0.013
	(0.239)	(0.438)	(0.031)	(0.008)
2	-0.774	0.213	-0.055	-0.027
	(0.386)	(0.739)	(0.064)	(0.013)
CZ-Year Trends	X	X	X	X
Bartiks	X	X	X	X
Y mean	6.71	58.40	2.74	3.19
Observations	5920	5920	5910	5920

Notes: Data are drawn from the 2005-2014 American Community Survey. The samples are based on working-age (20-64) low-educated (high school or less) foreign-born individuals and U.S.-born individuals. The models include CZ fixed effects, year fixed effects, CZ-specific linear trends and Bartik-style controls. Distributed lag model γ_k coefficients are estimated using equation (2) and then transformed into total treatment effects ($\beta_k s$), which are then reported, along with their standard errors, in the table. All results are weighted using the CZ population in 2000 and all standard errors are clustered by CZ. P-values not denoted.

Table A3
Effect of Secure Communities on Detentions

	Dep. Var: Total Detainers / Total CZ Pop in 2005 * 100	
	(1)	(2)
β: SC	0.121***	0.118***
	(0.021)	(0.021)
CZ-Year Trends	X	X
Bartiks		X
Y mean	0.10	0.10
% Effect	125.67	122.84
Observations	7400	7400

Notes: Data are from the 2005-2014 Transactional Records Access Clearinghouse (TRAC). Detentions are measured as total number of detentions in year t, divided by total working-age CZ Population in 2005, and then multiplied by 100. All models include CZ and year fixed effects and CZ-specific linear trends. The second column includes Bartik-style controls for local labor demand. The results are weighted using the CZ population in 2000. Standard errors are clustered by CZ and are reported in parentheses. * p<0.10, ** p<0.05, *** p<0.01

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Table A4

Effects of Secure Communities on the Unemployment and Labor Force Participation of Low-Educated Foreign-Born and U.S.-Born

	Total Group Unemp / Total Group Pop * 100		Total Group Out of LF / Total Group Pop * 100	
	LEFB	USB	LEFB	USB
β: SC	0.914***	0.363**	-0.264	0.034
	(0.304)	(0.174)	(0.229)	(0.062)
CZ-Year Trends	X	X	X	X
Bartiks	X	X	X	X
Y mean	5.14	5.13	7.99	9.23
% Effect	17.80	7.07	-3.30	0.37
Observations	7397	7400	7397	7400

Notes: Data are from the 2005-2014 American Community Survey. The sample is based on all working-age (20-64) individuals with subgroups denoted in the column headings. The unemployment share is measured as total unemployment in the demographic group by CZ and year, divided by total working-age CZ Population in 2005, and then multiplied by 100. The labor force share is measured as total number of out the labor force in the demographic group by CZ and year, divided by total working-age CZ Population in 2005, and then multiplied by 100. The models include CZ and year fixed effects, CZ-specific linear trends, and Bartik-style controls for labor demand. The results are weighted using the CZ population in 2000. Standard errors are clustered by CZ and are reported in parentheses. * p<0.10, ** p<0.05, *** p<0.01

Table A5

Effects of Secure Communities on Employment Share and Hourly Wages, by Citizenship Status

	Dep. Var: Total Group Emp / Total CZ Pop in 2005 * 100	
	Low-Educated Non-Citizens	Citizens
β: SC	-0.283***	-0.413**
	(0.079)	(0.173)
CZ-Year Trends	X	X
Bartiks	X	X
Y mean	4.37	64.62
% Effect	-6.48	-0.64
Observations	7400	7400
	Dep. Var: US-Born Log Hourly Wages	
	Low-Educated Non-Citizens	Citizens
β: SC	-0.013	-0.006***
	(0.013)	(0.002)
CZ-Year Trends	X	X
Bartiks	X	X
Y mean	2.63	3.18
Observations	7332	7400

Notes: Data are from the 2005-2014 American Community Survey. Column 1 reports the effects on working-age non-citizen individuals with a high school degree or less. Column 2 reports the effects on working-age U.S.-citizen individuals. The models include CZ and year fixed effects, CZ-specific linear trends, and Bartik-style controls for local labor demand. The results are weighted using the CZ population in 2000. Standard errors are clustered by CZ and are reported in parentheses. * p<0.10, ** p<0.05, *** p<0.01